

5.3.3 Activity

1. In 1991, the average consumption, in pounds, of nectarines was 6.88. For each year after 1991, the average consumption decreased 0.37 pounds per year. In what year, x , after 1991 will the average consumption be 3 pounds?

A

$$0.37x + 6.88 = 3$$

D

$$1991(3x - 0.37) = 6.88$$

B

$$6.88 - 0.37x = 3$$

E

$$3(6.88 - 0.37x) = 1991$$

C

$$3x + 1991 = 6.88 - 0.37$$

F

$$3(6.88 - 0.37x) = 1991$$

aa

The initial consumption in 1991.

rr

The rate at which consumption decreases.

kk

This value is not in the equation because it is informational.

ww

The consumption that is being solved for.

2. In 2010, the amount of money Floridians spent, in billions of dollars, on recreational goods such as video equipment and media, sporting equipment supplies, sports and recreational vehicles, and recreational books was \$19.116 billion. Each year since 2010 the amount of money Floridians spend has increased by \$1.352 billion. In what year, x , after 2010 did Floridians spend \$43.452 billion on recreational goods?

M

$$43.452x + 1.352 = 19.116$$

G

$$2010(1.352x + 19.116) = 43.452$$

L

$$1.352x + 19.116 = 43.452$$

H

$$1.352(19.116x + 43.452) = 2010$$

K

$$19.116x + 43.452 = 1.352$$

I

$$43.452(1.352x + 19.116) = 2010$$

mm

The amount Floridians spent in 2010.

tt

The rate at which spending increases.

bb

This value is not in the equation because it is informational.

hh

The amount Floridians spend that is being solved for.

3. Jessica is buying decorative rock for her garden. She is using a coupon that gives her the first 8 pounds for free. Each pound of decorative rock costs \$0.28 per pound. The total price Jessica paid is \$34.60, which includes \$2.40 in sales tax. How many pounds of rock, x , did Jessica get?

T $0.28x - 8 + 2.40 = 34.60$

P $(8x + .25) - 34.60 = 2.40$

S $0.28(x - 8 + 2.40) = 34.60$

R $8(34.60 - 0.28x) = 2.40$

W $0.28(x - 8) + 2.40 = 34.60$

V $8(0.28x - 34.60) = 2.40$

yy

The number of pounds of rock that was free.

gg

The cost of the rock per pound.

nn

The total price Jessica paid.

dd

The sales tax Jessica paid.